Supplementary Material

S1: Statistical Significance Testing

To ensure that the observed performance improvements are not caused by randomness in model training, we conduct statistical significance testing. Specifically, we adopt three random seeds (35, 42 [as reported in the original experiments], and 49) and repeat the training of each model under identical settings across the three affective conditions. Paired statistical tests (paired t-test) are then conducted based on the corresponding results. We design four comparisons: Base vs. +HIE, +HIE vs. +HIE+SSL, +HIE+SSL vs. +HIE+SSL+SLK, and +HIE+SSL+SLK vs. Full. The results (Table 1) demonstrate that the performance gains introduced by each module are statistically significant ($p < 0.05$), indicating that the observed improvements are stable and reliable rather than arising from random variation.

To enhance the rigor of the experiments, we further applied the Holm–Bonferroni procedure to correct for multiple comparisons. Specifically, for each stimulus condition and evaluation metric, corrected separately using the Holm–Bonferroni method. Pairwise comparisons between model configurations were conducted using results obtained under the same random seeds, thereby reducing the influence of stochastic variation during model training. As shown in Table 2, most of the performance improvements remained statistically significant after correction for multiple comparisons. Specifically, under the positive and neutral stimulus conditions, all comparisons for both Accuracy and F1 remained significant. Under the negative stimulus condition, all F1 comparisons, as well as the Accuracy comparison between +HIE+SSL+SLK and Full, remained statistically significant. However, the adjusted p-values for the first three incremental Accuracy comparisons under the negative stimulus condition were all 0.050, placing them at the conventional threshold of statistical significance.

Table 1: Statistical significance analysis of model components.

	Positive (P-V)		Neutral (P-V)		Negative (P-V)	
Comparison	Acc	F1	Acc	F1	Acc	F1
Base VS +HIE	0.015	0.012	0.037	0.035	0.018	0.017
+HIE VS +HIE+SSL	0.024	0.019	0.015	0.012	0.017	0.017
+HIE+SSL VS +HIE+SSL+SLK	0.022	0.021	0.018	0.015	0.018	0.016
+HIE+SSL+SLK VS Full	0.005	0.007	0.007	0.006	0.004	0.005

Base denotes the configuration without any additional components, while Full represents the complete model equipped with all components.

Table 2: Statistical significance analysis of model components (Holm-Bonferroni).

	Positive (P-V)		Neutral (P-V)		Negative (P-V)	
Comparison	Acc	F1	Acc	F1	Acc	F1
Base VS +HIE	0.045	0.036	0.045	0.036	0.050	0.048
+HIE VS +HIE+SSL	0.045	0.038	0.045	0.036	0.050	0.048
+HIE+SSL VS +HIE+SSL+SLK	0.045	0.038	0.045	0.036	0.050	0.048
+HIE+SSL+SLK VS Full	0.020	0.028	0.028	0.024	0.016	0.020

S2: Computational Resources Required for SFENet Deployment

The computational resources required for the overall deployment of SFENet are primarily associated with MTCNN and SFENet itself. We evaluated the resource requirements on an NVIDIA RTX 4090 GPU with 24 GB of memory. Because the modules are executed sequentially and operate on different computing devices, their peak resource demands do not occur simultaneously and therefore should not be directly summed. The quantitative results are presented in Table 3.

Table 3: Quantitative evaluation of the computational cost of SFENet deployment.

Processing Stage	Parameters	Computational Complexity	Peak GPU Memory	Average Inference Time

MTCNN	0.496 M	180.400 M MACs / 360.801 M FLOPs per frame	14.578 MiB	26.698 ms per frame
SFENet	0.197 M	7.877 M MACs / 15.753 M FLOPs per sample	1.194 MiB	15.912 ms per sample

S3: Comparison Between Model Interpretability Results and Clinical Expert

Annotationswe separately masked the model-identified high-response segments andthe expert-annotated segments. As shown in Table 4, model performance decreased in both cases, indicating that both types of segments contain important discriminative information and further supporting the validity of the clinical expert annotations. Notably, masking the model-identified high-response segments resulted in a more pronounced performance degradation, suggesting that the model may capture additional fine-grained information beyond that explicitly annotated by the experts. We further analyzed the AU composition and mean amplitudes of the expert-annotated and model-identified high-response segments. As shown in Table 4. Specifically, key action units, including AU04_r, AU10_r, AU25_r, AU05_r, and AU20_r, repeatedly appeared in both types of segments, demonstrating a strong correspondence between the facial movement patterns emphasized by the model and those identified by clinical experts. The differences were mainly confined to several lower-ranked AUs. For example, under the negative condition in the case group, the model placed greater emphasis on AU15_r, whereas the experts focused more on AU20_r. In the control group under the negative condition, the primary difference involved AU17_r and AU15_r. These findings suggest that the model is more sensitive to subtle tension-related changes in the chin and perioral muscles, whereas the experts tend to focus on lip-corner depression and lip stretching, which carry more explicit negative-emotion semantics. Overall, the two types of segments exhibit a high degree of similarity, while differing slightly in their emphasis on certain subtle facial-expression changes.

Regarding the mean amplitude, the model-identified segments exhibited values comparable to those of the expert-annotated segments and were slightly higher under most conditions, indicating that the model tended to localize temporal regions with more pronounced facial-expression changes. This difference may primarily arise from the distinct segment-selection mechanisms. The model identifies the most responsive temporal regions based on shapelet matching scores and is therefore more likely to select segments with concentrated AU activation. In contrast, the experts performed continuous annotation by jointly considering expression semantics, temporal dynamics, and contextual information; consequently, their annotations may encompass the onset, progression, and offset phases of facial expressions. Following further discussions with the two clinical experts, this interpretation was considered reasonable.

Table 3. Performance comparison under different types of segment masking.

EIVE/FAUs	Positive		Neutral		Negative	
Masking Type	Acc	F1	Acc	F1	Acc	F1
Model-Identified Segments	0.725	0.719	**0.668**	**0.665**	0.687	0.681
Expert-Annotated Segments	**0.731**	**0.727**	**0.668**	0.661	**0.691**	**0.689**

Table 4: Frequency of occurrence of the top five FAUs with the largest amplitude differences in EIVE. Here, "r" and "c" represent intensity and presence, respectively.

	Case group			Control group		
Sort	Positive	Neutral	Negative	Positive	Neutral	Negative
1	AU20_c (0.16)	AU10_r (0.14)	AU04_r (0.20)	AU12_r (0.21)	AU25_r (0.16)	AU04_r (0.20)
2	AU04_r	AU04_r	AU25_r	AU05_r	AU06_r	AU20_r

	(0.16)	(0.13)	(0.20)	(0.19)	(0.15)	(0.18)
3	AU05_r (0.14)	AU05_r (0.13)	AU10_r (0.18)	AU25_r (0.19)	AU05_r (0.15)	AU25_r (0.18)
4	AU17_r (0.13)	AU04_c (0.12)	AU20_r (0.17)	AU06_r (0.18)	AU26_c (0.14)	AU15_r (0.17)
5	AU12_r (0.13)	AU17_r (0.11)	AU04_c (0.16)	AU20_c (0.17)	AU10_c (0.12)	AU10_r (0.16)